

TrES-2b

R-2126

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120420 · created 2026-08-11T22:55:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456035.4213 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.023 +/- 0.044
Transit depth (Rp/Rs)^2	0.054 +/- 0.2 [%]
Orbital Inclination (inc)	83.35 +/- 2.91
Airmass coefficient 1 (a1)	1.0019 +/- 0.0074
Airmass coefficient 2 (a2)	-0.0016 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	0.74 %
Best Comparison Star	#1 - [248, 326]
Optimal Aperture	4.21
Optimal Annulus	9.47
Transit Duration (day)	0.029 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.023 ± 0.044 vs 0.1254 (deviation 1.8σ)
Duration fit vs published	0.029 d vs 0.0746 d (deviation 0.79σ)
Mid-time O–C	-3557.8 min
Depth SNR	0.4
Residual scatter	0.74 %

Light curve

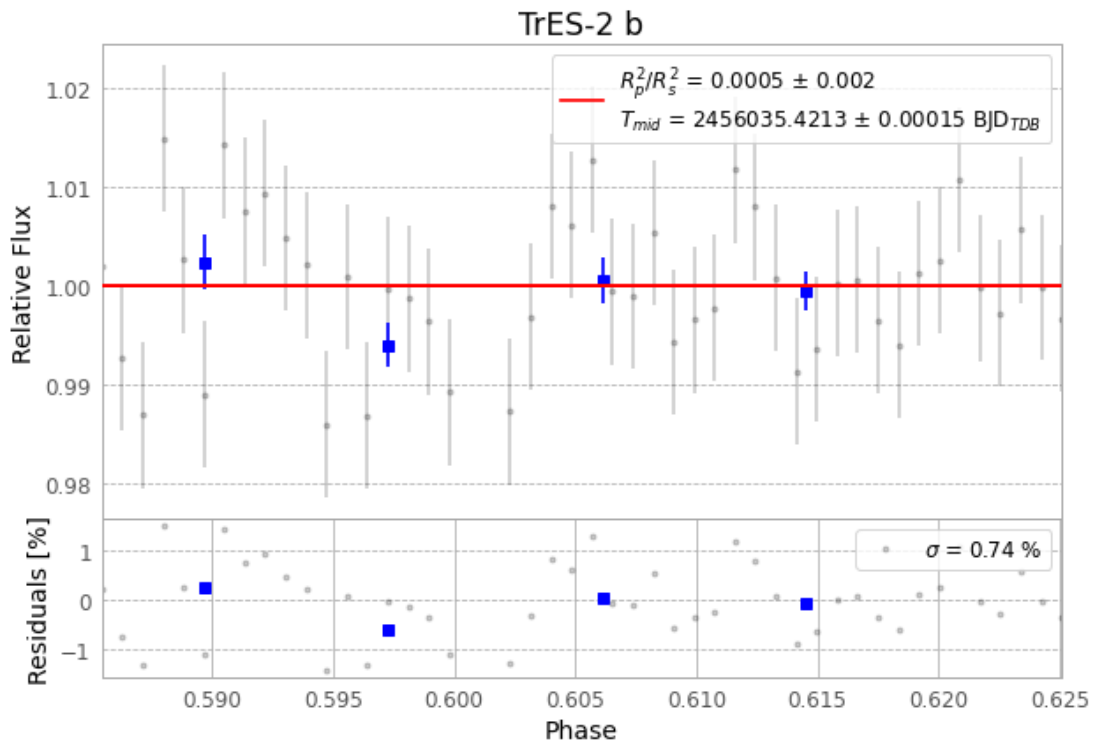


Figure 1 — FinalLightCurve_TrES-2 b_2012-04-19.png (SHA-256 fe37d7406e5256ed...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc062053a6491a81...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

46 FITS frames (30 MB), TRES-2120419084812.FITS → TRES-2120419110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120420.log (185f9577c879...), FinalLightCurve_TrES-2 b_2012-04-19.png (fe37d7406e52...), FinalLightCurve_TrES-2 b_2012-04-19.csv (e91e4ff9056a...)

Compute resources

Wall time 71.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 22:55Z.